

Detailed Syllabus - Artificial Intelligence (AI-101)

Course Overview

This course introduces the fundamental principles, methodologies, and applications of Artificial Intelligence. Students will explore intelligent agents, search strategies, knowledge representation, machine learning, neural networks, and emerging AI technologies with practical exposure.

Course Objectives

- Understand core concepts and foundations of Artificial Intelligence.
- Apply various search algorithms to solve real-world problems.
- Develop logical reasoning using knowledge representation techniques.
- Gain practical understanding of Machine Learning models.
- Analyze ethical and societal impacts of AI technologies.

UNIT I: Introduction to Artificial Intelligence (8 Hours)

- Definition and Scope of AI
- History and Evolution of AI
- Applications of AI in Industry and Society
- Intelligent Agents and Environment Types
- Problem Formulation and State Space Representation
- Uninformed Search Strategies: BFS, DFS, Uniform Cost Search
- Informed Search Strategies: Greedy Best First Search, A* Algorithm

UNIT II: Knowledge Representation and Reasoning (8 Hours)

- Propositional Logic and Predicate Logic
- Syntax and Semantics of First Order Logic
- Inference Rules and Resolution
- Forward and Backward Chaining
- Semantic Networks and Frames
- Introduction to Expert Systems and Rule-Based Systems

UNIT III: Machine Learning Fundamentals (10 Hours)

- Introduction to Machine Learning
- Supervised vs Unsupervised Learning
- Linear Regression and Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Trees and Random Forest
- Model Evaluation Metrics: Accuracy, Precision, Recall, F1-score
- Cross Validation Techniques

UNIT IV: Neural Networks and Deep Learning (10 Hours)

- Biological Neuron vs Artificial Neuron
- Perceptron Model and Learning Rule
- Multilayer Perceptron (MLP)
- Activation Functions
- Backpropagation Algorithm
- Introduction to Convolutional Neural Networks (CNN)
- Introduction to Recurrent Neural Networks (RNN)

UNIT V: Advanced AI Topics and Applications (8 Hours)

- Natural Language Processing Basics
- Computer Vision Fundamentals
- Reinforcement Learning Overview
- Robotics and Autonomous Systems
- AI in Healthcare, Finance, and Smart Cities
- Ethical Issues in AI: Bias, Privacy, Accountability

Laboratory Component (Practical)

- Implementation of BFS and DFS in Python
- Implementation of A* Search Algorithm
- Building a Rule-Based Expert System
- Linear Regression using Python
- Decision Tree Classifier Implementation
- Simple Neural Network using TensorFlow or PyTorch
- Mini Project: AI-based Application Development

Assessment Pattern

- Internal Assessment: 40% (Assignments, Quizzes, Lab Work)
- End Semester Examination: 60%
- Mini Project Evaluation included in Internal Assessment

Recommended Textbooks

- Stuart Russell & Peter Norvig – Artificial Intelligence: A Modern Approach
- Ethem Alpaydin – Introduction to Machine Learning
- Ian Goodfellow, Yoshua Bengio & Aaron Courville – Deep Learning